4-Bit Binary Arithmetic Calculator Using Combinational Logic

This project presents the design and implementation of a 4-bit binary arithmetic calculator built entirely using combinational logic, without any microprocessor or sequential control. The system performs addition and subtraction on two 4-bit binary numbers through a cascaded adder-subtractor circuit constructed from basic logic gates (AND, OR, XOR). Subtraction is achieved using the 2's complement method, where a mode-select control line activates XOR gates to conditionally invert the subtrahend before addition. The design was verified through truth tables, Boolean simplification, and circuit simulation, confirming that combinational logic can reliably perform fundamental arithmetic operations central to computer arithmetic units.

Key Points:

1. **Objective:** Design a 4-bit calculator that performs addition and subtraction using only combinational logic, without a microprocessor.
2. **Method:** A cascaded 4-bit adder-subtractor is built from AND, OR, and XOR gates, with 2's complement subtraction controlled by a single mode-select input.
3. **Verification:** Correctness is confirmed through truth tables, Boolean expression derivation, and simulation (e.g., Logisim/Multisim), with carry/borrow signals indicating overflow.
4. **Outcome:** The circuit achieves low-latency, gate-efficient arithmetic, illustrating the foundational logic behind ALUs in modern digital systems.

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